QUANTITATIVE FINANCE MASTER GUIDES

Black–Scholes and Option Greeks

An Interview-Oriented Mathematical and Economic Guide

PRICING	SENSITIVITY	TRADING
Derive the model	Understand the Greeks	Hedge and interpret

From terminal payoffs to the full risk surface

A rigorous, self-contained progression through no-arbitrage pricing, the Black–Scholes–Merton derivation, analytical Greeks, moneyness and maturity asymptotics, implied-volatility dynamics, delta/gamma hedging, and trading-interview problem solving.

Unifying learning path: Payoff → Distribution → Price → Derivatives → Hedge → P&L.

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How to Use This Guide

This guide has three complementary goals. First, derive the Black–Scholes–Merton (BSM) model without treating its formula as a magic rule. Second, understand every standard Greek as a derivative, an economic mechanism, and a hedge-management instruction. Third, recognize interview traps: ATM delta is not exactly one-half, higher implied volatility does not universally raise delta, and short maturity increases gamma mainly *near the strike*.

Four questions to ask on every options interview problem
1. Is the position a call or put, long or short? **2.** Is it ITM, ATM, or OTM?
3. Which inputs change, and which are held constant? **4.** Are we discussing model partial derivatives, or market movements where implied volatility also changes?

Scope and conventions. Unless noted, the underlying is an *ex-dividend stock or ETF price* $S > 0$, strike $K > 0$, calendar time t, expiry T, time to expiry $\tau = T - t > 0$, annualized volatility $\sigma > 0$, constant continuously compounded risk-free rate r, and continuous proportional dividend yield q. The baseline instrument is a *European vanilla option*. In practical equity/ETF option markets, contracts may be American-style and dividends may be discrete; those require adjustments discussed later. Φ and φ denote the standard normal CDF and PDF.

PART I Financial intuition and payoffs

1 Option payoffs, moneyness, and what a Greek measures

1.1 Payoffs and convexity

At expiry, a call gives the right to buy at K; a put gives the right to sell at K:

$$C_T = (S_T - K)^+ = \max(S_T - K, 0), \qquad P_T = (K - S_T)^+. \tag{1}$$

The payoff is *convex*: a call has slope 0 below K and 1 above K; a put has slope –1 below K and 0 above K. The slope at exactly $S_T = K$ is undefined. Before maturity, conditional uncertainty smooths these kinks into differentiable option-price curves.

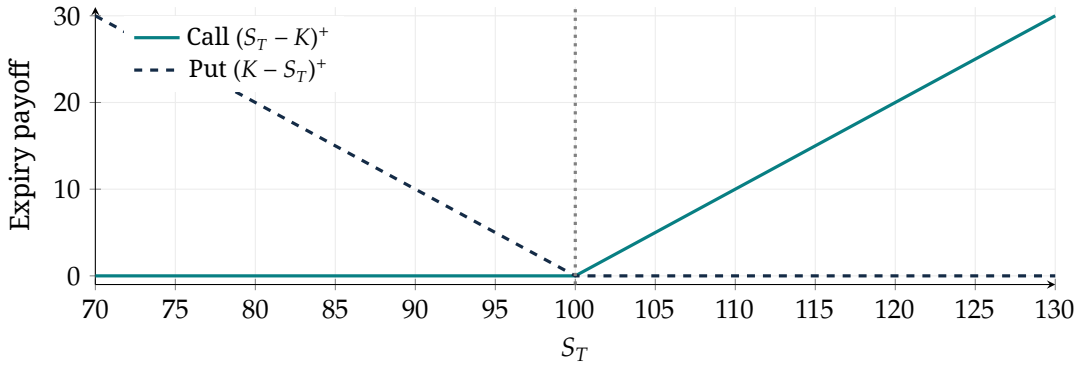


Figure 1: Vanilla payoffs with $K = 100$; premium paid at initiation is excluded.

1.2 Moneyness: definitions before memorizing signs

For a call: $S > K$ is ITM, $S = K$ is spot-ATM, and $S < K$ is OTM. For a put the ITM and OTM regions reverse. *Forward ATM* means $F_{t,T} = K$, where $F_{t,T} = Se^{(r-q)\tau}$; *delta-neutral ATM* refers to a strike chosen by a delta convention. These different ATM definitions need not coincide.

1.3 Net asset value versus underlying price

For a stock or ordinary ETF option, the variable S in BSM is the *tradable underlying share price*, not necessarily the ETF's reported NAV. ETF market price can deviate from net asset value. The option's own premium is V, not S, K, or NAV. For an option on a futures contract, the natural formulation is Black–76 rather than unmodified stock BSM; see section 13.

1.4 Greeks are partial derivatives, not unconditional predictions

Let $V(S, \sigma, t, r, q; K, T)$ be an option value. Then

$$\Delta = V_S, \quad \Gamma = V_{SS}, \quad \mathscr{V} = V_\sigma, \quad \Theta = V_t = -V_\tau, \quad \rho = V_r. \tag{2}$$

A *partial* derivative changes one parameter while holding the others fixed. In markets, S and implied volatility can move together; that gives a *total* sensitivity different from the textbook delta. All displayed Greeks refer to a *long one-unit option* unless a short position or contract multiplier is stated.

PART II Derive the Black–Scholes–Merton model

2 Model assumptions and no-arbitrage logic

2.1 The assumed stochastic price dynamics

The ex-dividend underlying price follows geometric Brownian motion with continuous dividend yield q:

$$\frac{\mathrm{d}S_t}{S_t} = (\mu - q)\mathrm{d}t + \sigma \mathrm{d}W_t, \tag{3}$$

where μ is the asset's total expected return under the physical measure. In the frictionless BSM setting: borrowing/lending at r is possible, markets permit continuous rehedging, volatility and rates are constant, and the underlying has continuous paths. There are no transaction costs, funding frictions, jumps, or early exercise in the European baseline.

What is being priced? A model option price is the cost of a self-financing trading strategy that replicates the contingent payoff, not simply a forecast of the option's average payoff under a trader's subjective probability.

2.2 Derivation I: Itô's lemma and a delta-hedged portfolio

Apply Itô's lemma to $V(S,t)$:

$$\mathrm{d}V = \left(V_t + (\mu - q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\mathrm{d}t + \sigma S V_S \mathrm{d}W. \tag{4}$$

Construct a portfolio long one option and short $\Delta = V_S$ shares of stock. After including the dividend stream $qS\Delta \mathrm{d}t$ on the short-stock leg, its instantaneous gain is

$$\mathrm{d}G = \mathrm{d}V - \Delta(\mathrm{d}S + qS\mathrm{d}t) \tag{5}$$

$$= \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - qSV_S\right)\mathrm{d}t. \tag{6}$$

The Brownian shock cancels. A locally riskless, self-financing portfolio with market value $V - SV_S$ must earn $r(V - SV_S)\mathrm{d}t$. Hence

$$\boxed{V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV = 0.} \tag{7}$$

This is the **Black–Scholes–Merton PDE**, subject to the terminal call or put payoff in eq. (1). The stock's physical drift μ has vanished: perfect local hedging eliminates the need to price the stock's expected return.

Do not omit dividends in the replication argument. A short-stock hedge owes the dividend stream. Omitting this cash flow would incorrectly produce the no-dividend PDE when $q \neq 0$.

2.3 Derivation II: risk-neutral terminal distribution

In the corresponding risk-neutral model, the stock's drift becomes $r - q$:

$$\frac{\mathrm{d}S_t}{S_t} = (r-q)\mathrm{d}t + \sigma \mathrm{d}W_t^{\mathbb{Q}}. \tag{8}$$

Integrating its log gives

$$S_T = S\exp\left[(r - q - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\,Z\right], \qquad Z \sim N(0,1). \tag{9}$$

The European call price is the discounted risk-neutral payoff:

$$C = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \tag{10}$$

$$= e^{-r\tau}\left\{\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T>K}] - K\mathbb{Q}(S_T > K)\right\}. \tag{11}$$

Define

$$\boxed{d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.} \tag{12}$$

Since $S_T > K$ exactly when $Z > -d_2$,

$$\mathbb{Q}(S_T > K) = \Phi(d_2). \tag{13}$$

For the truncated stock expectation, use completion of the square:

$$e^{az}\varphi(z) = e^{a^2/2}\varphi(z-a), \qquad \mathbb{E}[e^{aZ}\mathbf{1}_{Z>b}] = e^{a^2/2}\Phi(a-b). \tag{14}$$

With $a = \sigma\sqrt{\tau}$ and $b = -d_2$ this yields

$$\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T>K}] = Se^{(r-q)\tau}\Phi(d_1). \tag{15}$$

Substitute into eq. (10):

Black–Scholes–Merton closed forms

$$\boxed{C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),} \tag{BS–C}$$

$$\boxed{P = Ke^{-r\tau}\Phi(-d_2) - Se^{-q\tau}\Phi(-d_1).} \tag{BS–P}$$

The put formula follows by repeating the payoff integral or by using put–call parity below.

2.4 How to remember d1 and d2

Define the forward price $F = Se^{(r-q)\tau}$. Then

$$d_1 = \frac{\ln(F/K)}{\sigma\sqrt{\tau}} + \frac{\sigma\sqrt{\tau}}{2}, \qquad d_2 = \frac{\ln(F/K)}{\sigma\sqrt{\tau}} - \frac{\sigma\sqrt{\tau}}{2}. \tag{16}$$

The essential scale is *total volatility* $\sigma\sqrt{\tau}$: a normalized distance of log-forward price from strike. Recall "*forward moneyness divided by total volatility, plus/minus half of total volatility*."

2.5 The difference between Phi(d1) and Phi(d2)

By eq. (13), $\Phi(d_2)$ is the *risk-neutral probability* that a no-dividend European call finishes ITM. By contrast $\Phi(d_1)$ is the no-dividend *spot delta.* Delta changes the initial stock price and weights favorable

high-stock outcomes more heavily than an indicator probability. Neither should be casually equated with a real-world probability under the physical measure.

2.6 Put–call parity and the Greeks of parity

Subtract put value from call value:

$$\boxed{C - P = Se^{-q\tau} - Ke^{-r\tau}.} \tag{17}$$

Economic replication: a long call and short put at common K, T replicate a prepaid forward. Differentiating parity gives

$$\boxed{\Delta_C - \Delta_P = e^{-q\tau}, \quad \Gamma_C = \Gamma_P, \quad \mathscr{V}_C = \mathscr{V}_P.} \tag{18}$$

In addition, $\Theta_C - \Theta_P = qSe^{-q\tau} - rKe^{-r\tau}$ and $\rho_C - \rho_P = K\tau e^{-r\tau}$. These are fast consistency checks in an interview.

PART III Calculate and interpret every Greek

3 Delta: price slope and hedge ratio

3.1 A direct derivation from the call formula

The key differentiation identity is

$$\boxed{Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2).} \tag{19}$$

It follows from $d_1 = d_2 + \sigma\sqrt{\tau}$ and the normal-density ratio $\varphi(d_1)/\varphi(d_2) = \exp[-(d_1^2 - d_2^2)/2]$. Differentiate $C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$ with respect to S:

$$\begin{aligned}\frac{\partial C}{\partial S} &= e^{-q\tau}\Phi(d_1) + Se^{-q\tau}\varphi(d_1)\frac{\partial d_1}{\partial S} - Ke^{-r\tau}\varphi(d_2)\frac{\partial d_2}{\partial S} \\ &= e^{-q\tau}\Phi(d_1),\end{aligned}$$

where the derivative terms cancel by eq. (19) and $\partial d_1/\partial S = \partial d_2/\partial S = 1/(S\sigma\sqrt{\tau})$. From parity:

$$\boxed{\Delta_C = e^{-q\tau}\Phi(d_1), \qquad \Delta_P = e^{-q\tau}[\Phi(d_1) - 1] = -e^{-q\tau}\Phi(-d_1).} \tag{20}$$

For $q = 0$, the delta of a long call lies between 0 and 1, and the delta of a long put between –1 and 0. For dividend-paying underlyings, magnitudes are multiplied by $e^{-q\tau}$, which can make the simple zero-dividend bounds inaccurate as literal formulas.

3.2 Economic meaning and hedging units

A delta of 0.60 means that a *small* \$1 rise in S raises option value by approximately \$0.60, holding IV and other model inputs constant. If a call contract represents 100 ETF shares and a dealer is short 100 contracts with delta 0.50, the dealer has roughly $-100\times100\times0.5 = -5{,}000$ shares of stock-equivalent delta and can initially buy about 5,000 shares to offset it.

3.3 Why spot-ATM delta approaches one-half

Set $S = K$, $r = q = 0$. Then $d_1 = \sigma\sqrt{\tau}/2$ and

$$\Delta_C = \Phi(\sigma\sqrt{\tau}/2) \underset{\tau\downarrow 0}{\longrightarrow} \Phi(0) = \frac{1}{2}. \tag{21}$$

The call payoff's slope is 0 below strike and 1 above strike; just before expiry at spot-ATM, the tiny remaining price uncertainty puts nearly equal mass on either side. More precisely, Δ is $\Phi(d_1)$, *not* the ITM probability $\Phi(d_2)$. At expiry and exactly $S_T = K$, the kink makes the payoff derivative undefined; 1/2 is the pre-expiry limit.

ATM $\Delta_C = 0.5$ is an approximation, not an identity. With $S = K, r = q = 0$ and $\sigma = 20\%$, call deltas are approximately 0.5021 at 1 day, 0.5114 at 30 days, and 0.5398 at one year. At forward-ATM ($F = K$), $d_1 = \sigma\sqrt{\tau}/2$ and the call delta is $e^{-q\tau}\Phi(\sigma\sqrt{\tau}/2)$, which also need not equal 0.5.

3.4 Terminal delta limits

For fixed $S \neq K$ and vanishing time, assuming finite rates and volatility,

$$\Delta_C \to \begin{cases} 0 & S < K \\ 1 & S > K \end{cases}, \qquad \Delta_P \to \begin{cases} -1 & S < K \\ 0 & S > K \end{cases}. \tag{22}$$

This provides a memorable mental image: near maturity the delta curve becomes a step function around the strike. The *speed of this transition* is gamma.

4 Gamma: how quickly delta changes with spot

4.1 Derivation and meaning

Gamma is the second derivative of the option value with respect to S:

$$\Gamma = \frac{\partial \Delta}{\partial S} = V_{SS}. \tag{23}$$

Differentiate eq. (20) using $\Phi'(x) = \varphi(x)$:

$$\boxed{\Gamma_C = \Gamma_P = \frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}} > 0.} \tag{24}$$

If delta is 0.50 and gamma is 0.04 per dollar, a \$1 increase in underlying price moves delta to approximately 0.54 and a \$1 decrease moves it to about 0.46; only a local approximation, since gamma changes during the move.

4.2 Why gamma concentrates around the strike

Far OTM call delta is already near 0; far ITM call delta is near 1. Both change slowly with the underlying. Near the exercise boundary delta transitions quickly and the price curve bends: gamma is large. Long calls *and* long puts have positive gamma. Short calls/puts have negative position gamma.

4.3 Time and volatility asymptotics

At $S = K$ in the short-maturity limit, $d_1 \to 0$ and $e^{-q\tau} \to 1$. Thus

$$\Gamma_{\text{ATM}} \sim \frac{1}{S\sigma\sqrt{2\pi\tau}}. \tag{25}$$

Hence gamma near the strike grows like $\tau^{-1/2}$ as expiry approaches, but for *fixed* $S \neq K$, $|d_1| \to \infty$ and the normal density $\varphi(d_1)$ decays faster than $\tau^{-1/2}$ grows: $\Gamma \to 0$. With spot approximately at strike, higher σ spreads the delta transition over a wider price range, normally reducing local gamma; for distant strikes, volatility can increase gamma instead.

A single visualization remembers three rules. Short-dated ATM: delta changes sharply across the strike, gamma is high. Short-dated deep ITM/OTM: delta is almost pinned at its final slope, gamma tends to zero. Higher volatility blurs the strike boundary and generally lowers peak near-ATM gamma.

5 Vega: price sensitivity to volatility

5.1 Derive vega from the pricing formula

Let $\mathscr{V} = V_\sigma$. By differentiating d_1 and d_2 in the call price and applying eq. (19), the derivative terms combine to yield

$$\boxed{\mathscr{V}_C = \mathscr{V}_P = Se^{-q\tau}\varphi(d_1)\sqrt{\tau} > 0.} \tag{26}$$

For an explicit intermediate check,

$$\frac{\partial d_1}{\partial \sigma} = -\frac{d_2}{\sigma}, \qquad \frac{\partial d_2}{\partial \sigma} = -\frac{d_1}{\sigma}. \tag{27}$$

Then $\partial C/\partial\sigma = Se^{-q\tau}\varphi(d_1)(\partial d_1/\partial\sigma - \partial d_2/\partial\sigma) = Se^{-q\tau}\varphi(d_1)\sqrt{\tau}$.

5.2 Economic logic: convex payoff values uncertainty

Greater risk-neutral volatility distributes terminal prices more widely. A long call benefits from large upward outcomes but cannot have a negative terminal payoff; a long put benefits from extreme downward outcomes while its payoff floor is zero. The convexity of either vanilla payoff makes model vega positive. *Positive vega does not mean that vega itself rises when volatility rises*: that depends on volga.

5.3 Vega units, at-the-money behavior, and maturity

The formula in eq. (26) measures premium change for a change of 1.00 in decimal volatility, e.g. 20% → 120%. For a one-volatility-point (20% → 21%) move, divide it by 100. Near ATM and expiry, $\varphi(d_1) \to \varphi(0)$:

$$\mathscr{V}_{\text{ATM}} \sim \frac{S\sqrt{\tau}}{\sqrt{2\pi}}. \tag{28}$$

Consequently the *ATM* vega tends to zero as $\tau \downarrow 0$ and generally grows with maturity over ordinary horizons. Vega is largest *near* ATM but its exact peak depends on whether spot or strike is varied and on carry. Deep OTM and deep ITM options have weak vega close to maturity because the payoff outcome becomes almost certain.

5.4 Exact gamma–vega identity

Divide eq. (26) by eq. (24):

$$\boxed{\mathscr{V} = \Gamma S^2 \sigma \tau.} \tag{29}$$

Memory: gamma and vega share the same normal-density term, but gamma scales as $1/\sqrt{\tau}$ while vega scales as $\sqrt{\tau}$ near ATM. "*Short-dated ATM: high gamma, low vega; long-dated ATM: lower gamma, larger vega.*"

6 Theta: the passage of calendar time

6.1 The sign convention

Time to expiry is $\tau = T - t$; as calendar time advances, τ falls. Market convention uses

$$\boxed{\Theta = \frac{\partial V}{\partial t} = -\frac{\partial V}{\partial \tau}.} \tag{30}$$

Thus a theta of -0.04 per day means that, if other risk inputs are held constant, passage of one short day costs about four cents in option value. It is a local derivative, not an exact one-day repricing for very short-dated options.

6.2 Full call and put theta formulas

Differentiate the BSM prices with respect to calendar time, or use the PDE with the other Greeks. For a continuous dividend yield:

$$\boxed{\Theta_C = -\frac{Se^{-q\tau}\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2) + qSe^{-q\tau}\Phi(d_1),} \tag{Theta–C}$$

$$\boxed{\Theta_P = -\frac{Se^{-q\tau}\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}\Phi(-d_2) - qSe^{-q\tau}\Phi(-d_1).} \tag{Theta–P}$$

In particular, with $r = q = 0$:

$$\boxed{\Theta_C = \Theta_P = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}}.} \tag{31}$$

This is negative for both calls and puts. For nonzero financing/carry, *it is not universally true that every long option has negative theta.* A deep ITM European put with positive rates may have positive theta.

6.3 Why ATM theta steepens near maturity

From eq. (31), ATM short-time theta is approximately

$$\Theta_{\text{ATM}} \sim -\frac{S\sigma}{2\sqrt{2\pi\tau}}. \tag{32}$$

A near-expiry ATM option rapidly loses the remaining value of uncertainty. For fixed OTM or ITM locations away from strike and $r = q = 0$, theta instead tends to zero as expiry approaches. With nonzero rates, deep ITM options can retain nonzero carry contributions near expiry.

6.4 Theta–gamma relation and the cost of convexity

Rewrite the PDE in Greeks:

$$\boxed{\Theta + \tfrac{1}{2}\sigma^2 S^2\Gamma + (r-q)S\Delta - rV = 0.} \tag{33}$$

Only if $r = q = 0$ does this reduce to

$$\boxed{\Theta = -\tfrac{1}{2}\sigma^2 S^2\Gamma.} \tag{34}$$

Economic meaning: long gamma buys convexity but (typically) pays negative theta; short gamma often receives time decay while absorbing adverse large-price-movement risk. It is a

risk/reward relationship, not a guaranteed profit rule.

7 Rho, dividend sensitivity, and strike sensitivity

7.1 Rho: interest-rate sensitivity

Define $\rho = \partial V/\partial r$. Differentiating BSM gives

$$\boxed{\rho_C = K\tau e^{-r\tau}\Phi(d_2) > 0, \qquad \rho_P = -K\tau e^{-r\tau}\Phi(-d_2) < 0.} \tag{35}$$

A higher discount rate decreases the present value of the future strike payment, generally benefiting European calls and hurting puts, holding spot and the remaining inputs fixed. Rho's magnitude is often more important for longer tenors, but its maturity dependence is not universally monotonic.

7.2 Dividend yield / carry sensitivity

Let $\psi = \partial V/\partial q$ with q treated as a continuous dividend yield. Then

$$\boxed{\psi_C = -\tau S e^{-q\tau}\Phi(d_1) < 0, \qquad \psi_P = \tau S e^{-q\tau}\Phi(-d_1) > 0.} \tag{36}$$

Higher expected dividends reduce the prepaid forward $Se^{-q\tau}$: calls fall and puts rise, everything else fixed. For discrete cash dividends and American ETF options, the treatment is more complicated than a single constant q.

7.3 How strike K changes the option price

Differentiating the payoff price with respect to K:

$$\boxed{\frac{\partial C}{\partial K} = -e^{-r\tau}\Phi(d_2) < 0, \qquad \frac{\partial P}{\partial K} = e^{-r\tau}\Phi(-d_2) > 0.} \tag{37}$$

A higher exercise price makes a call less useful but a put more valuable. Both call and put *delta* fall as strike rises, since

$$\boxed{\frac{\partial \Delta_C}{\partial K} = \frac{\partial \Delta_P}{\partial K} = -\frac{e^{-q\tau}\varphi(d_1)}{K\sigma\sqrt{\tau}} < 0.} \tag{38}$$

8 Higher-order Greeks: vanna, volga, charm, and speed

8.1 Vanna: how volatility changes delta

Since $\Delta = e^{-q\tau}\Phi(d_1)$ for calls, and the put differs by a σ-independent term,

$$\boxed{\text{Vanna} = \frac{\partial \Delta}{\partial \sigma} = -e^{-q\tau}\varphi(d_1)\frac{d_2}{\sigma}.} \tag{39}$$

Thus $\text{sign}(\text{Vanna}) = -\text{sign}(d_2)$. Higher IV *raises* delta if $d_2 < 0$, *lowers* it if $d_2 > 0$, and has zero first-order effect if $d_2 = 0$. This is more precise than the oversimplified statement "IV drives delta toward 0.5." Calls and puts of the same strike/maturity have identical vanna in BSM.

8.2 Volga / vomma: how volatility changes vega

Differentiate eq. (26) with respect to σ:

$$\boxed{\text{Volga} = \frac{\partial^2 V}{\partial \sigma^2} = \mathscr{V}\cdot\frac{d_1 d_2}{\sigma}.} \tag{40}$$

Vega is positive for a long European vanilla, but *volga may be positive or negative*. That is why increasing IV increases option price without necessarily increasing its vega.

8.3 Charm: time decay of delta

Define calendar-time charm as $\partial\Delta/\partial t$, at fixed S, K, σ, r, q:

$$\frac{\partial d_1}{\partial \tau} = \frac{(r - q + \sigma^2/2)\tau - \ln(S/K)}{2\sigma\tau^{3/2}}, \tag{41}$$

$$\text{Charm}_C = \frac{\partial \Delta_C}{\partial t} = \boxed{q e^{-q\tau}\Phi(d_1) - e^{-q\tau}\varphi(d_1)\frac{\partial d_1}{\partial \tau}}, \tag{Charm–C}$$

$$\text{Charm}_P = \frac{\partial \Delta_P}{\partial t} = \boxed{-q e^{-q\tau}\Phi(-d_1) - e^{-q\tau}\varphi(d_1)\frac{\partial d_1}{\partial \tau}} \tag{Charm–P}$$

Some desks define charm as $\partial\Delta/\partial\tau$ instead; that reverses its sign. Always ask which time convention is used. Economic meaning: delta can drift toward its terminal value even with no price movement.

8.4 Speed: how spot changes gamma

For completeness, the third spot derivative is

$$\boxed{\text{Speed} = \frac{\partial \Gamma}{\partial S} = -\frac{\Gamma}{S}\left(1 + \frac{d_1}{\sigma\sqrt{\tau}}\right).} \tag{42}$$

Speed warns that using a constant gamma for large underlying jumps may substantially misestimate delta hedges. Other higher-order risks include zomma ($\partial\Gamma/\partial\sigma$), color ($\partial\Gamma/\partial t$), and ultima ($\partial$ Volga $/\partial\sigma$); they are mostly desk- and product-specific rather than first-round memorization priorities.

PART IV How every Greek changes when inputs change

9 The systematic sensitivity map

The correct order of reasoning is: *identify long/short and call/put; locate moneyness; choose the changing input; state which other inputs remain fixed; then differentiate or take a limit*. Never infer a second derivative's sign solely from the first derivative's sign.

9.1 Changes in the underlying price S

At fixed K, σ, r, q, τ:

- **Call price rises** because $\Delta_C > 0$; **put price falls** because $\Delta_P < 0$.
- **Call and put deltas both rise** because $\Gamma_C = \Gamma_P > 0$. Put delta rises from near –1 to near 0; it does not become more negative.
- **Gamma is strongest near the strike** (not mathematically always exactly $S = K$); gamma is small far ITM/OTM for ordinary vol/maturity values.
- **Vega is strongest near the exercise boundary**; its exact spot location need not equal K.
- **Theta and vanna have no universal change direction** across all moneyness and carry regimes.

In fact, differentiating log-gamma with respect to S yields eq. (42); at fixed K, τ, σ, r, q, the gamma maximum as a *function of spot* occurs where $d_1 = -\sigma\sqrt{\tau}$, hence $S = K \exp[-(r - q + 3\sigma^2/2)\tau]$. It lies near, not identically at, spot-ATM when maturity is small. The vega maximum as a function of spot instead satisfies $d_1 = \sigma\sqrt{\tau}$. These distinctions matter for a mathematically exact interview answer, not for the basic rule "near ATM."

9.2 Changes in strike K

At fixed underlying spot, expiry, and volatility:

- Raising K reduces call price and increases put price, by eq. (37).
- Raising K decreases both call and put deltas ($\partial\Delta/\partial K < 0$).
- Gamma and vega are generally largest for strikes near the at-the-money region rather than monotonically increasing or decreasing across all strikes.

When varying strike at fixed S, τ, σ, r, q, vega is maximal at $d_1 = 0$ (because its prefactor $Se^{-q\tau}\sqrt{\tau}$ is held fixed); changing *spot* instead produces a different optimum. Which quantity is varied must be stated.

9.3 Changes in implied volatility sigma

Object	Effect of increasing IV, with other BSM inputs held constant
Call and put values	Both strictly rise: vega is positive.
Call/put delta	Can rise or fall. The *exact local sign* of vanna is opposite to that of d_2.
Gamma	Typically falls near ATM when IV increases; may rise far away from the strike. No universal direction across moneyness.
Vega itself	Can increase or decrease: determined by volga = $\mathscr{V} d_1 d_2/\sigma$.
Theta	Under $r = q = 0$, depends jointly on σ and d_1; no universal monotone response across moneyness.

For example, with $r = q = 0, K = 100, \tau = 0.25$ years:

Call moneyness	Δ_C at IV 20%	Δ_C at IV 40%	Direction
$S = 95$ (OTM)	0.322	0.438	up
$S = 100$ (ATM)	0.520	0.540	up slightly
$S = 105$ (ITM)	0.705	0.635	down

Wrong shortcut: "higher vol pushes every delta toward 0.5." For a zero-carry spot-ATM call, delta is $\Phi(\sigma\sqrt{\tau}/2)$ and *increases* as IV rises. The exact direction for any moneyness follows $-d_2$, not the vague notion of moving toward a half. For extreme IV even a spot-ITM call's vanna can change sign.

9.4 Changes in time to maturity tau

For options *near the strike* as $\tau \downarrow 0$ and r, q remain finite:

$$\Gamma \sim \frac{1}{S\sigma\sqrt{2\pi\tau}}, \qquad \mathscr{V} \sim \frac{S\sqrt{\tau}}{\sqrt{2\pi}}, \qquad \Theta_{r=q=0} \sim -\frac{S\sigma}{2\sqrt{2\pi\tau}}. \tag{43}$$

This gives the standard interview rule: **short-dated ATM = high gamma, low vega, rapidly negative theta**. For S held strictly away from K, gamma and vega tend to zero as expiry approaches; delta tends to the slope of the terminal payoff. Theta's limit in deep ITM contracts can include r and q carry and need not vanish.

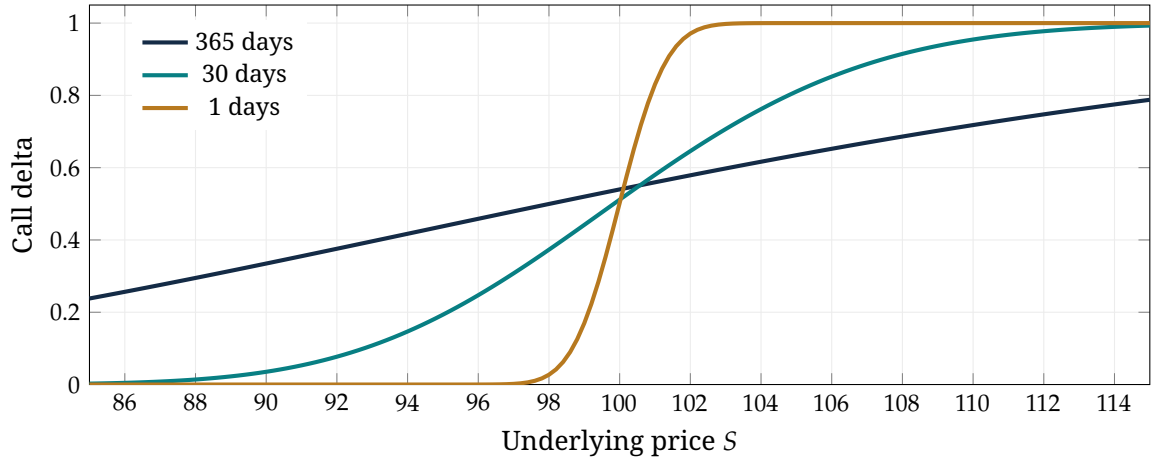


Figure 2: Black–Scholes call delta for $K = 100$, $\sigma = 20\%$, $r = q = 0$, and three expiries. Shorter maturity creates an increasingly sharp transition around the strike.

9.5 Time to maturity need not change delta monotonically

The often-heard claim "call delta increases as expiry approaches" is false in general. With $q = 0$, the delta is $\Phi(d_1)$, where

$$\boxed{\frac{\partial d_1}{\partial \tau} = \frac{(r - q + \sigma^2/2)\tau - \ln(S/K)}{2\sigma\tau^{3/2}}.} \tag{44}$$

This sign can be positive or negative. Near expiry, a fixed ITM call delta tends to 1 and OTM call delta tends to 0, but *far from expiry* the path need not be monotonic if parameters change or carry matters. Spot-ATM $r = q = 0$ delta tends to 0.5 from above as time vanishes. For puts, the $q = 0$ delta difference is always one, so $\partial\Delta_P/\partial\tau = \partial\Delta_C/\partial\tau$.

9.6 Rates and dividend yield

Holding S, K, σ, τ, q fixed, raising r increases European call value, decreases put value, and *increases both call and put deltas*:

$$\frac{\partial \Delta_C}{\partial r} = \frac{\partial \Delta_P}{\partial r} = e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma} > 0. \tag{45}$$

A higher continuous dividend yield q decreases call value and increases put value, but delta sensitivity needs care:

$$\frac{\partial \Delta_C}{\partial q} = -\tau e^{-q\tau}\Phi(d_1) - e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma} < 0, \tag{46}$$

$$\frac{\partial \Delta_P}{\partial q} = \tau e^{-q\tau}\Phi(-d_1) - e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma}, \tag{47}$$

whose sign need not be universal for puts. Gamma, vega, theta, and higher-order sensitivities to r or q generally vary with strike, maturity, and the discount factors; avoid unsupported all-purpose arrows.

9.7 Compact directional map, with explicit qualifiers

Changed input	Call effect	Put effect	Important qualification
$S \uparrow$	Price up; delta up	Price down; delta up (less negative)	Gamma is positive for long vanilla calls and puts.
$K \uparrow$	Price down; delta down	Price up; delta down (more negative)	At fixed S, σ, τ, r, q.
$\sigma \uparrow$	Price up; delta ambiguous	Price up; delta ambiguous	Delta sign controlled by vanna; gamma/vega changes may be ambiguous.
$\tau \downarrow$	Delta $\to 0, 1$, or ATM ≈ 0.5	Delta $\to -1, 0$, or ATM ≈ -0.5	Near-strike gamma rises; far-strike gamma falls. ATM vega falls.
$r \uparrow$	Price up; delta up	Price down; delta up	European BSM at fixed spot and dividend yield.
$q \uparrow$	Price down; delta down	Price up; delta ambiguous	Continuous proportional dividend yield.

PART V Market practice: volatility surfaces and risk

10 Implied volatility, realized volatility, and path-dependent markets

10.1 What is implied volatility?

Realized volatility describes movements observed in historical or subsequent price data; a common estimator is the annualized standard deviation of log returns. *Implied volatility* is the σ that makes a chosen option pricing model match a given market option premium, all other inputs specified. In practice IV reflects risk premia, supply/demand, jump risks, and the market's forward-looking uncertainty; it is not mechanically determined by recent realized volatility.

10.2 One volatility for all strikes is often unrealistic

Real option markets exhibit *skew* or *smile*: $\sigma_{\text{imp}}(K,\tau)$ varies with strike and expiry. Two options with different strikes need not have the same IV even when other BSM inputs match. A dealer's option risk therefore includes volatility *level*, *skew*, *term structure*, and the dynamics of their changes.

10.3 Sticky strike versus sticky delta

Sticky strike assumes IV at a fixed absolute strike remains unchanged under a small change in S (at fixed maturity). Then

$$\frac{\mathrm{d}\sigma_{\text{imp}}(K,\tau)}{\mathrm{d}S} = 0 \quad \Rightarrow \quad \Delta_{\text{total}} = \Delta_{\text{BS}}. \tag{48}$$

Sticky delta assumes IV is fixed along a particular delta/moneyness label as the underlying moves. Because the delta label associated with a *fixed strike* changes when S changes, that strike's implied volatility generally moves. Under any specified smile-dynamics rule, the price sensitivity is

$$\boxed{\Delta_{\text{total}} = \Delta_{\text{BS}} + \mathscr{V}\frac{\mathrm{d}\sigma_{\text{imp}}}{\mathrm{d}S}.} \tag{49}$$

This is a *total derivative along the assumed market-volatility response*, not the pure BSM partial delta. Neither sticky strike nor sticky delta is a universal law. A jump may change skew, term structure, correlation assumptions, and smile dynamics in ways not captured by a single convention.

10.4 How delta moves when spot, IV, and time all change

For small joint movements,

$$\boxed{\mathrm{d}\Delta \approx \Gamma\,\mathrm{d}S + \text{Vanna}\;\mathrm{d}\sigma + \text{Charm}\;\mathrm{d}t + \frac{\partial\Delta}{\partial r}\mathrm{d}r + \frac{\partial\Delta}{\partial q}\mathrm{d}q.} \tag{50}$$

The first term is gamma-driven spot sensitivity; the second is vanna-driven vol sensitivity; the third is calendar time sensitivity. A large stock jump may require still higher-order corrections: these are local derivatives, not finite-change exact formulas.

11 Interview case: two price paths end at the same spot

Question. A stock/ETF is flat for weeks and suddenly jumps; alternatively, it declines gradually and then rallies by the same final amount. In both cases it finishes at the same current price. Which scenario has higher option delta? Explain in pure BSM and with market IV dynamics.

Layer 1: theoretical answer. At the comparison instant, if S, K, τ, r, q, σ are identical, $\Delta = e^{-q\tau}\Phi(d_1)$ for a call in either path. Historical prices do not appear in the model's state variables. **Thus the deltas are equal.** If the final jump amount is also the same, the pre-jump spots are equal; if pre-jump IV/time inputs are also equal, the pre-jump BSM deltas are equal as well.

Layer 2: real markets. A surprising jump after quiet trading might lead to a sharp IV reprice; a prolonged selloff may already have elevated IV, while a rebound can trigger a volatility crush. These are conditional market scenarios, not guaranteed laws: sometimes a jump resolves uncertainty and IV falls. Without the actual IV response, strike, maturity, and smile conventions, *one cannot rank the deltas*.

Layer 3: use vanna rather than slogans. If scenario A has higher IV than B, for a forward-OTM call with $d_2 < 0$ delta will typically be higher in A locally; for sufficiently ITM options with $d_2 > 0$ delta moves lower with IV. At spot-ATM $r = q = 0$, raising IV increases $\Phi(\sigma\sqrt{\tau}/2)$ slightly. The statement "higher IV always pushes delta to 0.5" is false.

Layer 4: dealer action. The desk must update the delta hedge using gamma and vanna, assess the implied-volatility surface, and account for discontinuous price moves that defeat frictionless instantaneous hedging.

Interview answer in one sentence. *Same current inputs imply identical BSM delta; different market IV surfaces can produce different deltas, but the ordering cannot be determined from price path alone, and the IV-to-delta sign depends on vanna/moneyness.*

12 Option P&L as a Greeks expansion

Let small changes over a short interval be $\Delta S, \Delta\sigma, \Delta t, \Delta r$. A second-order Taylor expansion begins with

$$\Delta V \approx \Delta\,\Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \mathscr{V}\,\Delta\sigma + \Theta\,\Delta t + \rho\,\Delta r \\ + \text{Vanna}(\Delta S)(\Delta\sigma) + \frac{1}{2}\,\text{Volga}(\Delta\sigma)^2 + \cdots. \tag{51}$$

Where the same glyph Δ denotes both the Greek and a finite change, read $\Delta\,\Delta S$ as "delta (Greek) times change in S." Cross terms with time and rates, and higher powers of price changes, are omitted. This is an *approximate local risk attribution*; large jumps and surface repricing warrant a full-model revaluation.

12.1 Dealer hedge example: short 100 calls

A dealer sells 100 call contracts, each controlling 100 ETF shares, with per-share $\Delta_C = 0.50$ and $\Gamma = 0.04$ per \$1 of underlying price change. The desk's stock-equivalent delta is $-100 \times 100 \times 0.50 = -5{,}000$. Buy 5,000 underlying shares to be initially delta-neutral. If the ETF rises by \$1, the call delta approximately rises to 0.54. The short-option position delta becomes $-5{,}400$. The 5,000 shares remain +5,000 delta, leaving -400 net. **Buy roughly 400 more shares** to restore neutrality. If the ETF falls \$1, call delta falls and the short-gamma desk must sell stock instead.

Hedge-flow memory rule
Short gamma: buy more after price rises; sell after price falls (unfavorable buy-high/sell-low convexity, before accounting for premiums/theta).
Long gamma: sell some hedge shares after price rises; buy them back after price falls (favorable sell-high/buy-low rebalancing, but carry negative theta).

12.2 Why gamma profits and theta losses are linked

Start with a long option and continuously offset its instantaneous delta with stock. In a zero-interest, no-dividend, very short-horizon approximation, with IV held fixed:

$$\Delta\Pi_{\text{delta hedged}} \approx \frac{1}{2}\Gamma(\Delta S)^2 + \Theta\Delta t. \tag{52}$$

For example with $\Gamma = 0.07$, daily $\Theta = -0.038$, and a \$2 move over the interval, the naive frozen-Greeks approximation gives $\frac{1}{2}(0.07)(2)^2 - 0.038 \simeq \$0.102$ per underlying unit. If the underlying barely moves, theta may dominate. This illustration neglects rehedging timing, changes in gamma, transaction costs, and IV repricing.

12.3 A more precise realized-versus-implied variance statement

Price a continuously delta-hedged long European option using an implied BSM volatility σ_{imp}. If the realized stock process is continuous with instantaneous quadratic-variation rate σ^2_{real}, interest/dividend financing is handled consistently, and rehedging is idealized as continuous and frictionless, excess hedged gain locally satisfies

$$\boxed{\mathrm{d}\Pi \simeq \frac{1}{2}\Gamma S^2\left(\sigma^2_{\text{real}} - \sigma^2_{\text{imp}}\right)\mathrm{d}t.} \tag{53}$$

Intuitively, *long gamma benefits from realized variance exceeding the variance paid for through option pricing*; short gamma often benefits from the reverse, all else equal. This is not a universal profit guarantee: smile changes, jumps, costs, discrete hedging, financing, vega exposure, and the path of gamma affect realized desk P&L. It is a conditional relationship under a highly stylized hedge model.

13 Options on ETFs, dividends, and futures

An ordinary ETF option references ETF *share price* S, not necessarily NAV. Its distributions may affect forward price and early-exercise decisions. US-listed equity/ETF options are often American-style, so early exercise can matter, especially near ex-dividend dates for calls and with rates/dividends for puts. BSM European formulas are a fundamental baseline, not a universal exchange settlement or pricing rule.

For options on a *futures contract* with current forward/futures level F and expiry τ, the commonly used Black–76 formulas are

$$\begin{aligned} C_F &= e^{-r\tau}\left[F\Phi(\tilde{d}_1) - K\Phi(\tilde{d}_2)\right], \\ P_F &= e^{-r\tau}\left[K\Phi(-\tilde{d}_2) - F\Phi(-\tilde{d}_1)\right], \\ \tilde{d}_1 &= \frac{\ln(F/K) + \sigma^2\tau/2}{\sigma\sqrt{\tau}}, \qquad \tilde{d}_2 = \tilde{d}_1 - \sigma\sqrt{\tau}. \end{aligned} \tag{54}$$

Here $\partial C_F/\partial F = e^{-r\tau}\Phi(\tilde{d}_1)$ is a *futures delta*, not the same as stock BSM spot delta if the forward depends

on the spot through cost of carry. BSM and Black–76 agree after the appropriate change of underlying and carry conventions.

PART VI Interview problems and systematic solution methods

14 A six-step method for any Greeks question

Step 1. Identify the instrument. Call or put? European, American, or futures option? Long or short? What is the contract multiplier?

Step 2. Classify moneyness. Spot ATM or forward ATM? ITM or OTM? Is the distance from strike large relative to $\sigma\sqrt{\tau}$?

Step 3. Hold the right inputs fixed. Is this a partial derivative under one IV, or a joint price/volatility/smile move? Is maturity short or long?

Step 4. Choose the minimal mathematical identity. Use $\Delta = e^{-q\tau}\Phi(d_1)$; $\Gamma = \Delta_S$; $\mathscr{V} = \Gamma S^2\sigma\tau$; $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma + (r-q)S\Delta - rV = 0$; vanna if IV changes delta.

Step 5. Infer sign, limiting behavior, and exception. Ask if the rule is global, near-ATM, or near-expiry. Check the relevant first or second derivative.

Step 6. Translate into a hedge. Determine position delta/gamma, what shares or other options the desk trades, and what risk is left unhedged.

15 Fully worked interview questions

15.1 Why does a spot-ATM call have delta about one-half?

Reasoning. The payoff slope at expiry jumps from 0 (OTM) to 1 (ITM). Before expiry, uncertainty smooths the change. With $S = K, r = q = 0$, $d_1 = \sigma\sqrt{\tau}/2$, so $\Delta_C = \Phi(\sigma\sqrt{\tau}/2) \to 0.5$ as $\tau \to 0$. Put delta approaches –0.5.
Answer. Only approximately 0.5 near expiry, not at every maturity. At exact expiry and strike the terminal payoff derivative does not exist.

15.2 Is call delta the probability of finishing in the money?

No. In BSM, with zero dividends, call delta is $\Phi(d_1)$ while the risk-neutral ITM probability is $\Phi(d_2)$. They differ by $d_1 - d_2 = \sigma\sqrt{\tau}$. Higher terminal stock outcomes have a larger weight in the delta derivative than in a simple exercise indicator. Only near the at-the-money, near-expiry limit do both approach one-half.

15.3 What happens to gamma as maturity approaches zero?

First ask where the strike is. Near ATM, $\Gamma \sim 1/(S\sigma\sqrt{2\pi\tau}) \to \infty$; at fixed $S \neq K$, $\Gamma \to 0$. The reason is that the delta curve approaches the stepwise slope of the option's terminal payoff and becomes steep only at the strike.

15.4 Why does Vega vanish as expiry approaches?

BSM vega is $Se^{-q\tau}\varphi(d_1)\sqrt{\tau}$. For ATM options its magnitude is of order $\sqrt{\tau}$, so it tends to zero. There is almost no remaining time for a change in annualized implied volatility to alter the terminal payoff distribution. Fixed ITM/OTM options also have vega tending to zero.

15.5 Which has more risk: near-expiry gamma or long-dated vega?

They are *different* risks, not a simple ordering. An ATM near-expiry option may have enormous gamma but very small vega; a long-dated ATM option typically has greater vega but a smoother delta response to S. State the unit of risk and the anticipated market movement before comparing economic exposures.

15.6 Does IV doubling always increase call delta?

No. Call option *price* increases because vega is positive, but the *delta* response is vanna: $-e^{-q\tau}\varphi(d_1)d_2/\sigma$. It is positive when $d_2 < 0$, negative when $d_2 > 0$. With $K = 100, \tau = .25, r = q = 0$, increasing IV from 20% to 40% raises delta for $S = 95$ (approximately $0.322 \to 0.438$), but lowers delta for $S = 105$ ($0.705 \to 0.635$).

15.7 Does a volatility rise always increase vega?

No. Positive *vega* means the option value rises when IV rises. Whether vega rises is determined by volga, $\partial\mathscr{V}/\partial\sigma = \mathscr{V}d_1d_2/\sigma$, which can have either sign. Distinguish a Greek from its own derivative.

15.8 How can positive gamma and negative theta both be rational?

Gamma is positive convexity: a delta-hedged long option can benefit from large underlying movements. Theta is the premium paid in passing time. When $r = q = 0$, $\Theta = -\frac{1}{2}\sigma^2S^2\Gamma$. A desk buys option convexity if the hedging opportunities/realized volatility justify that theta and the transaction/hedging costs.

15.9 If long/short gamma, how does a delta-neutral hedge trade after an up move?

Suppose a dealer is short calls and initially delta-neutral by owning shares. A rise in S raises the calls' delta (positive *option* gamma), making the *short* option position more negative delta. The dealer buys more stock to restore neutrality. On a fall, it sells stock. Long gamma requires the reverse. A larger gamma implies more shares must be traded per unit stock move.

15.10 Hedging 100 short call contracts with known delta and gamma

Initially option delta is $-100 \cdot 100 \cdot 0.5 = -5{,}000$: buy 5,000 shares. After a \$1 spot rise and assuming gamma approximately constant, delta per option becomes 0.54, hence total short-option delta $-5{,}400$. Buy roughly 400 *additional* shares. The calculations are first-order in delta and may change under IV shocks.

15.11 Why is a put's delta negative, and why does put delta increase with spot?

The put payoff is $\max(K - S_T, 0)$, so a higher underlying price reduces its value: negative delta. But its payoff is convex, so gamma is positive; put delta becomes *less negative* when spot rises (for instance $-0.7 \to -0.4$). At zero dividends, put delta equals call delta minus one.

15.12 Are call and put gamma/vega equal?

For matched European vanilla options with the same underlying, K, T, σ, r, q, yes. Differentiate put–call parity twice with respect to S for gamma and once with respect to σ for vega. But their deltas differ by $e^{-q\tau}$, and rho/theta need not match.

15.13 What happens to a call when the risk-free rate rises?

At fixed spot, volatility, expiry, and dividend yield, the European call becomes more valuable and the put less valuable. The strike's discounted payment $Ke^{-r\tau}$ falls. Call rho is $K\tau e^{-r\tau}\Phi(d_2) > 0$ and put rho is negative. A higher r also increases d_1 and hence both call and put deltas.

15.14 How do expected ETF dividends affect calls and puts?

A higher continuous dividend yield reduces the prepaid forward $Se^{-q\tau}$: call values fall, put values rise. Mathematically $\partial C/\partial q = -\tau S e^{-q\tau}\Phi(d_1) < 0$; $\partial P/\partial q = \tau S e^{-q\tau}\Phi(-d_1) > 0$. American exercise before ex-dividend dates and discrete distributions require a different treatment from fixed-yield European BSM.

15.15 Two historical paths finish at the same stock price. Which has higher delta?

For identical current BSM inputs, the deltas are identical regardless of path. In a real market, paths can influence implied vol and skew; delta can then differ because σ differs and the total delta includes the surface response. Higher IV does not universally mean higher delta; one needs strike, tenor, current IVs, and vanna. Thus no unconditional ranking can be drawn from the two paths alone.

15.16 Why is sticky-delta delta not necessarily BSM delta?

BSM delta is $\partial V/\partial S$ at constant σ. Under a sticky-delta or other smile-movement assumption, IV at fixed strike changes when the underlying moves. Apply the chain rule: $\mathrm{d}V/\mathrm{d}S = \Delta_{\mathrm{BS}} + \mathscr{V}\,\mathrm{d}\sigma_{\mathrm{imp}}/\mathrm{d}S$. State the surface convention and quote units.

15.17 Why can a short straddle be profitable with positive theta despite negative gamma?

A short ATM straddle is short a call and put, so it tends to have approximately offsetting initial deltas, negative gamma, negative vega, and positive theta under standard settings. It can profit from time decay if realized movement and IV changes are sufficiently small relative to the premium collected. Large market moves can overwhelm the theta income; initial near-zero delta does not imply low risk.

15.18 Explain IV-versus-realized-volatility in a delta-hedged option trade.

A stylized zero-friction, continuously delta-hedged long option has local excess gain $\frac{1}{2}\Gamma S^2(\sigma^2_{\mathrm{real}} - \sigma^2_{\mathrm{imp}})\mathrm{d}t$. Long gamma can benefit when realized variance exceeds the variance implied by its purchase price. This is conditional on the path, hedge quality and changes in the vol surface; it is not a guaranteed arbitrage.

15.19 How do you distinguish delta, gamma, and vanna in one sudden jump?

For option *price*: delta multiplies ΔS, gamma contributes $\frac{1}{2}\Gamma(\Delta S)^2$, and vega multiplies $\Delta\sigma$. For the *new delta*: a local approximation is $\Delta_{\mathrm{new}} \approx \Delta_{\mathrm{old}} + \Gamma\Delta S + \mathrm{Vanna}\,\Delta\sigma + \mathrm{Charm}\,\Delta t$. If the jump is large, reprice the full model because Greeks change.

15.20 If the ETF price is unchanged but time passes, can the option delta change?

Yes. Delta depends on time to maturity via d_1, even if spot is unchanged. This effect is charm. ITM/OTM deltas generally drift toward their terminal limits, while ATM $r = q = 0$ call delta approaches 0.5. If IV also changes, vanna contributes independently.

16 A full trader's answer to the motivating interview question

Spoken answer (roughly 55–75 seconds).
"In the Black–Scholes model, delta depends on today's stock price, strike, volatility, time to maturity, rates, and dividends, but not directly on the historical price path. So if the two scenarios finish at the same price and all current inputs are the same, the deltas must be identical. In practice the paths may change implied volatility and the volatility surface. A sudden jump after

quiet trading may raise volatility, while a rebound after a selloff could reduce it, but neither reaction is guaranteed. The effect on delta depends on moneyness: the sign of vanna is opposite to the sign of d_2. I would therefore avoid choosing a higher delta without knowing the strike and updated implied volatility. As a market maker, I would reprice the options and update the delta hedge for both spot and volatility movements."

17 Numerical reference: maturity and moneyness

At $S = K = 100$, $r = q = 0$, $\sigma = 20\%$, with a calendar year of 365 days:

Remaining expiry	Call delta	Gamma	Vega per +1 IV pt	Theta per day
1 day	0.5021	0.3811	\$0.0209	-\$0.2088
30 days	0.5114	0.0695	\$0.1143	-\$0.0381
365 days	0.5398	0.0198	\$0.3970	-\$0.0109

The values are *per single underlying unit*. Vega is given for a $+0.01$ (one percentage-point) change in annualized volatility, and theta is the instantaneous annual-rate derivative divided by 365. Actual contract P&L must apply the option multiplier, position sign, and risk recalculation after finite changes.

APPENDIX A Complete BSM formula and sensitivity atlas

A Black–Scholes and Option Greeks: Complete Reference

How to use this appendix. It is a stand-alone, interview-speed reference for the entire preceding guide. Read in this order: **model/notation → prices → Greek formulas → exact derivatives → limiting cases → directional maps → trading applications**. Every arrow holds the *other BSM inputs fixed*, unless a different market-volatility convention is explicitly stated. "ATM" means *spot-ATM* ($S = K$) unless qualified. All contracts below are *European*; Greek numbers refer to *one long unit*.

A.1 Definitions, units, and signs

Symbol	Meaning / unit	Always specify in an interview
S, K, V	Spot (ETF share) price, strike, option premium	S is the traded underlying price, *not* automatically an ETF's NAV; $V = C$ or P.
$t, T, \tau = T - t$	Calendar time, expiry, years left	Theta $V_t = -V_\tau$; years, not trading-day counts, enter BSM.
r, q, σ	Continuously compounded rate, continuous dividend yield, annual vol	All rates and volatility in *decimal* units (20% means 0.20).
$F = Se^{(r-q)\tau}$	Forward price	Spot-ATM $S = K$ and forward-ATM $F = K$ are different if $r \neq q$.
$\Phi(x), \varphi(x)$	Standard-normal CDF and PDF	$\Phi' = \varphi$; $\varphi'(x) = -x\varphi(x)$; $\varphi(0) = 1/\sqrt{2\pi}$.
Δ, Γ	V_S, V_{SS}	Per \$1 in spot; short positions reverse the signs.
$\mathscr{V}, \Theta, \rho$	V_σ, V_t, V_r	Vega/1 vol point $= 0.01\mathscr{V}$; daily theta $= \Theta/365$; rho per 1 rate point $= 0.01\rho$.
M, n	Contract multiplier, signed contracts	Risk and P&L $= nM\times$ per-unit Greek. For a short position, $n < 0$.

A.2 The complete Black–Scholes pricing engine

Object	Formula and interpretation
Payoffs	$C_T = (S_T - K)^+$; $P_T = (K - S_T)^+$. Both are convex; the payoff slope has a kink at $S_T = K$.
Physical dynamics	$\mathrm{d}S_t/S_t = (\mu - q)\mathrm{d}t + \sigma \mathrm{d}W_t$; μ is the physical expected *total* return and cancels from the replication argument.
Risk-neutral dynamics	$\dfrac{\mathrm{d}S_t}{S_t} = (r - q)\mathrm{d}t + \sigma \mathrm{d}W_t^{\mathbb{Q}}$.
Terminal distribution	$S_T = S\exp[(r - q - \frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z]$, $Z \sim N(0,1)$.
Risk-neutral value	$V = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[\text{payoff}(S_T)]$.
Forward and prepaid forward	$F = Se^{(r-q)\tau}$; $Se^{-q\tau} = Fe^{-r\tau}$.
Total vol and moneyness	$w = \sigma\sqrt{\tau}$, $x = \ln(F/K)/w$. Therefore $d_1 = x + w/2$, $d_2 = x - w/2$.
Direct definitions	$d_1 = \dfrac{\ln(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}$, $d_2 = d_1 - \sigma\sqrt{\tau}$.
CDF / PDF	$\Phi(x) = \displaystyle\int_{-\infty}^{x} \frac{e^{-u^2/2}}{\sqrt{2\pi}}\,\mathrm{d}u$, $\varphi(x) = \dfrac{e^{-x^2/2}}{\sqrt{2\pi}}$.
European call	$\boxed{C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)}$.
European put	$\boxed{P = Ke^{-r\tau}\Phi(-d_2) - Se^{-q\tau}\Phi(-d_1)}$.
Put–call parity	$\boxed{C - P = Se^{-q\tau} - Ke^{-r\tau}}$.
Density cancellation	$\boxed{Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)}$. The fastest route to all simple Greek formulas.
ITM risk-neutral probability	$\mathbb{Q}(S_T > K) = \Phi(d_2)$; call delta instead uses $\Phi(d_1)$ (for $q = 0$).
BSM PDE	$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV = 0}$. Boundary at expiry is the corresponding payoff.

The shortest derivation map. Apply Itô to $V(S,t)$ → short V_S units of stock against one long option → cancel the $\mathrm{d}W$ term → require riskless return r → obtain the PDE → solve by the discounted risk-neutral payoff. In the closed form, the $S_T > K$ probability produces $\Phi(d_2)$ and the stock-weighted truncated expectation produces $\Phi(d_1)$.

A.3 First-order Greeks: full call–put table

Read this as definition → closed form → economic exposure. Carry terms are retained; therefore these are more general than zero-dividend shortcuts.

Greek	Call	Put	Intuition / sign
Delta V_S	$e^{-q\tau}\Phi(d_1)$	$-e^{-q\tau}\Phi(-d_1)$	Call +, put –; stock-equivalent hedge.
Gamma V_{SS}	$\frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}}$	*Identical to call*	Positive for both long options; curve convexity.
Vega V_σ	$Se^{-q\tau}\varphi(d_1)\sqrt{\tau}$	*Identical to call*	Positive for both; vol-uncertainty value.
Theta V_t	$-\frac{Se^{-q\tau}\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2) + qSe^{-q\tau}\Phi(d_1)$	$-\frac{Se^{-q\tau}\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}\Phi(-d_2) - qSe^{-q\tau}\Phi(-d_1)$	Usually negative for long options; carry can alter sign.
Rho V_r	$K\tau e^{-r\tau}\Phi(d_2)$	$-K\tau e^{-r\tau}\Phi(-d_2)$	Call +, put – for fixed q.
Dividend V_q	$-\tau Se^{-q\tau}\Phi(d_1)$	$+\tau Se^{-q\tau}\Phi(-d_1)$	Call –, put + for larger q.
Strike V_K	$-e^{-r\tau}\Phi(d_2)$	$+e^{-r\tau}\Phi(-d_2)$	Call –, put + as K rises.

The three reconstruction identities

$$\boxed{\Gamma = \frac{\partial \Delta}{\partial S}}, \qquad \boxed{\mathscr{V} = S^2\sigma\tau\Gamma}, \qquad \boxed{\Theta + \tfrac{1}{2}\sigma^2 S^2\Gamma + (r-q)S\Delta - rV = 0}.$$

The final formula becomes $\Theta = -\frac{1}{2}\sigma^2 S^2\Gamma$ *only* at $r = q = 0$. This explains the long-gamma / negative-theta trade-off; it does not imply a guaranteed option-trading profit.

A.4 Higher-order Greeks and exact derivatives for change-of-Greeks questions

First memorize the d_1, d_2 derivatives; they eliminate almost all rote learning.

$$\frac{\partial d_1}{\partial S} = \frac{1}{S\sigma\sqrt{\tau}}, \qquad \frac{\partial d_1}{\partial K} = -\frac{1}{K\sigma\sqrt{\tau}}, \qquad \frac{\partial d_1}{\partial r} = \frac{\sqrt{\tau}}{\sigma}, \qquad \frac{\partial d_1}{\partial q} = -\frac{\sqrt{\tau}}{\sigma},$$

$$\frac{\partial d_1}{\partial \sigma} = -\frac{d_2}{\sigma}, \qquad \frac{\partial d_2}{\partial \sigma} = -\frac{d_1}{\sigma}, \qquad \frac{\partial d_1}{\partial \tau} = \frac{(r-q+\sigma^2/2)\tau - \ln(S/K)}{2\sigma\tau^{3/2}}.$$

Define $a_\tau := \partial d_1/\partial\tau$ (so $\partial_\tau d_2 = a_\tau - \sigma/(2\sqrt{\tau})$); all time derivatives below fix S, K, σ, r, q.

Name	Exact derivative (European BSM)	Interpretation / sign

Name	Exact derivative (European BSM)	Interpretation / sign
Vanna	$V_{S\sigma} = \frac{\partial \Delta}{\partial \sigma} = \frac{\partial \mathscr{V}}{\partial S} = -e^{-q\tau}\varphi(d_1)\frac{d_2}{\sigma}$	Sign = $-\operatorname{sign}(d_2)$; IV may raise or lower delta.
Volga / vomma	$V_{\sigma\sigma} = \frac{\partial \mathscr{V}}{\partial \sigma} = \mathscr{V}\frac{d_1 d_2}{\sigma}$	Sign = $\operatorname{sign}(d_1 d_2)$; vega need not rise with IV.
Ultima	$V_{\sigma\sigma\sigma} = \frac{\mathscr{V}}{\sigma^2}\left(d_1^2 d_2^2 - d_1 d_2 - d_1^2 - d_2^2\right)$	Third derivative in implied volatility; sign not fixed.
Charm (call)	$\frac{\partial \Delta_C}{\partial t} = qe^{-q\tau}\Phi(d_1) - e^{-q\tau}\varphi(d_1)a_\tau$	Calendar-time convention; some desks use $\partial_\tau \Delta$.
Charm (put)	$\frac{\partial \Delta_P}{\partial t} = -qe^{-q\tau}\Phi(-d_1) - e^{-q\tau}\varphi(d_1)a_\tau$	Delta can drift without spot moving.
Speed	$\frac{\partial \Gamma}{\partial S} = -\frac{\Gamma}{S}\left(1 + \frac{d_1}{\sigma\sqrt{\tau}}\right)$	Gamma changes with spot; no universal sign.
Zomma	$\frac{\partial \Gamma}{\partial \sigma} = \Gamma\frac{d_1 d_2 - 1}{\sigma}$	ATM commonly negative; away from money sign can reverse.
Color	$\frac{\partial \Gamma}{\partial t} = \Gamma\left(q + \frac{1}{2\tau} + d_1 a_\tau\right)$	Calendar-time color; beware sign convention.
Vega / maturity	$\frac{\partial \mathscr{V}}{\partial \tau} = \mathscr{V}\left(-q + \frac{1}{2\tau} - d_1 a_\tau\right)$	ATM vega scales as $\sqrt{\tau}$ near expiry.
Gamma / maturity	$\frac{\partial \Gamma}{\partial \tau} = \Gamma\left(-q - \frac{1}{2\tau} - d_1 a_\tau\right)$	Opposite of calendar-time color.
Strike-of-delta	$\frac{\partial \Delta_{C,P}}{\partial K} = -\frac{e^{-q\tau}\varphi(d_1)}{K\sigma\sqrt{\tau}} < 0$	Both call and put delta decline as strike rises.
Rate-of-delta	$\frac{\partial \Delta_{C,P}}{\partial r} = e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma} > 0$	Both deltas increase with r.
Dividend-of-delta	$\frac{\partial \Delta_C}{\partial q} = -\tau e^{-q\tau}\Phi(d_1) - e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma}$	Call delta decreases as q rises.
Dividend-of-put-delta	$\frac{\partial \Delta_P}{\partial q} = \tau e^{-q\tau}\Phi(-d_1) - e^{-q\tau}\varphi(d_1)\frac{\sqrt{\tau}}{\sigma}$	Can be either sign.
Time-of-delta	$\frac{\partial \Delta_{C,P}}{\partial \tau} = \begin{cases} -qe^{-q\tau}\Phi(d_1) + e^{-q\tau}\varphi(d_1)a_\tau, & C \\ qe^{-q\tau}\Phi(-d_1) + e^{-q\tau}\varphi(d_1)a_\tau, & P \end{cases}$	Negative of corresponding calendar-time Charm.

Further cross-Greek derivative (theta versus IV). Differentiate the Greek-form BSM PDE:

$$\boxed{\frac{\partial \Theta}{\partial \sigma} = -\sigma S^2\Gamma - \frac{1}{2}\sigma^2 S^2\,\text{Zomma} - (r-q)S\,\text{Vanna} + r\mathscr{V}.}$$

No universal sign follows: do not say that higher IV must increase or decrease theta without specifying other parameters.

A.5 Option and Greek changes when one input rises: interview arrow matrix

Reading rule: ↑ means increasing the column's named parameter while keeping all other BSM inputs, including fixed *spot or strike*, unchanged; ± means no universal sign. For $\tau \downarrow$, use the separate limit tables that follow.

Quantity	$S\uparrow$	$K\uparrow$	$\sigma\uparrow$	$r\uparrow$	$q\uparrow$
Call premium C	↑	↓	↑	↑	↓
Put premium P	↓	↑	↑	↓	↑
Call delta Δ_C	↑	↓	± (Vanna)	↑	↓
Put delta Δ_P	↑	↓	± (Vanna)	↑	±
Gamma Γ	± (Speed)	±	± (Zomma)	±	±
Vega $\mathscr{V}$	± (Vanna)	±	± (Volga)	±	±
Theta Θ	±	±	±	±	±
Rho ρ	↑	±	±	±	±

Exact criteria behind the ± entries. Use Vanna $\propto -d_2$ for $\partial_\sigma \Delta$ and $\partial_S \mathscr{V}$; Volga $\propto d_1 d_2$ for $\partial_\sigma \mathscr{V}$; Zomma $\propto d_1 d_2 - 1$ for $\partial_\sigma \Gamma$; and Speed $\propto -(1 + d_1/(\sigma\sqrt{\tau}))$ for $\partial_S \Gamma$. They provide a *proof* where arrows alone cannot.

A.6 Moneyness map: long call versus long put

Fixed spot region	**Call delta,** $\tau \downarrow 0$	**Put delta,** $\tau \downarrow 0$	**Gamma and vega near expiry**
$S < K$ (call OTM; put ITM)	0	–1	Both $\to$ 0 super-polynomially in $1/\tau$.
$S = K$ (spot ATM)	1/2	–1/2	$\Gamma \to \infty$ as $\tau^{-1/2}$; $\mathscr{V} \to 0$ as $\tau^{1/2}$.
$S > K$ (call ITM; put OTM)	1	0	Both $\to$ 0 super-polynomially in $1/\tau$.

Extra theta limit at fixed non-ATM spot. When $S > K$, $\Theta_C \to qS - rK$ and $\Theta_P \to 0$; when $S < K$, $\Theta_P \to rK - qS$ and $\Theta_C \to 0$. These are financing/dividend *carry terms*: with $r = q = 0$, all four non-ATM thetas tend to zero.

The ATM point is special. At expiry itself and $S_T = K$, the *payoff derivative is undefined.* The values +1/2, –1/2 are pre-expiry limits. At fixed $S \neq K$, the limits of Γ and $\mathscr{V}$ are zero, not infinity. The term "OTM" or "ITM" must be held *fixed as* $\tau \to 0$ for these rows to apply.

A.7 Near-expiry asymptotic formula table: exactly when and what is proportional to what

Conditions: $S = K > 0$ *held fixed*, $\sigma > 0$, finite constant r, q, and $\tau \downarrow 0$. Symbol $f \sim g$ means $f/g \to 1$; Θ's leading term also holds with nonzero finite r, q.

Quantity	Leading expansion / limit at spot ATM	Scaling
d_1, d_2	$d_{1,2} = \dfrac{r - q \pm \sigma^2/2}{\sigma}\sqrt{\tau}$	$\propto \sqrt{\tau}$
Call delta	$\dfrac{1}{2} + \dfrac{r - q + \sigma^2/2}{\sigma\sqrt{2\pi}}\sqrt{\tau} + O(\tau)$	$\Delta_C \to 1/2$
Put delta	$-\dfrac{1}{2} + \dfrac{r - q + \sigma^2/2}{\sigma\sqrt{2\pi}}\sqrt{\tau} + O(\tau)$	$\Delta_P \to -1/2$
Gamma	$\Gamma \sim \dfrac{1}{S\sigma\sqrt{2\pi\tau}}$	$\propto \tau^{-1/2}\sigma^{-1}$
Vega	$\mathscr{V} \sim \dfrac{S\sqrt{\tau}}{\sqrt{2\pi}}$	$\propto \tau^{1/2}$
Theta (call and put)	$\Theta \sim -\dfrac{S\sigma}{2\sqrt{2\pi\tau}}$	$\propto -\tau^{-1/2}$
Rho (call, put)	$\rho_C \sim +K\tau/2; \quad \rho_P \sim -K\tau/2$	$\propto \pm\tau$
Dividend sensitivity	$\psi_C \sim -S\tau/2; \quad \psi_P \sim +S\tau/2$	$\propto \mp\tau$
Vanna ($r = q = 0$)	$\text{Vanna} \sim \dfrac{\sqrt{\tau}}{2\sqrt{2\pi}}$	$\propto \tau^{1/2}$
Volga ($r = q = 0$)	$\text{Volga} \sim -\dfrac{S\sigma\tau^{3/2}}{4\sqrt{2\pi}}$	$\propto -\tau^{3/2}$
Charm ($r = q = 0$)	$\text{Charm} \sim -\dfrac{\sigma}{4\sqrt{2\pi\tau}}$	$\propto -\tau^{-1/2}$

For *fixed* $S \neq K$ instead, $|d_1| \sim |\ln(S/K)|/(\sigma\sqrt{\tau}) \to \infty$ and $\varphi(d_1) = O(e^{-c/\tau})$ for some $c > 0$. Thus Γ, $\mathscr{V}$ and their higher spot/vol derivatives tend to zero, despite their explicit powers of τ in denominators.

A.8 Other limiting regimes: volatility, deep moneyness, and long tenor

Limit / condition	BSM price and delta	Gamma / vega and useful caveats

Limit / condition	BSM price and delta	Gamma / vega and useful caveats
$\sigma \downarrow 0$, fixed τ, $F > K$	$C \to e^{-r\tau}(F-K)$, $\Delta_C \to e^{-q\tau}$; put $\to 0$.	$\Gamma, \mathscr{V} \to 0$ away from $F = K$; payoff becomes deterministic.
$\sigma \downarrow 0$, fixed τ, $F < K$	$P \to e^{-r\tau}(K-F)$, $\Delta_P \to -e^{-q\tau}$; call $\to 0$.	$\Gamma, \mathscr{V} \to 0$ away from $F = K$.
$\sigma \downarrow 0$, $F = K$	$C, P \to 0$, $\Delta_C \to \frac{1}{2}e^{-q\tau}$, $\Delta_P \to -\frac{1}{2}e^{-q\tau}$.	$\Gamma \sim e^{-q\tau}/(S\sigma\sqrt{2\pi\tau}) \to \infty$; $\mathscr{V} \to Se^{-q\tau}\sqrt{\tau}/\sqrt{2\pi}$.
$\sigma \to \infty$, fixed τ	$C \to Se^{-q\tau}$, $P \to Ke^{-r\tau}$; $\Delta_C \to e^{-q\tau}$, $\Delta_P \to 0$.	$\Gamma, \mathscr{V} \to 0$. *Delta does not universally tend to 0.5.*
$S \downarrow 0$, fixed K, τ	$C \to 0$, $P \to Ke^{-r\tau}$; $\Delta_C \to 0$, $\Delta_P \to -e^{-q\tau}$.	$\Gamma, \mathscr{V} \to 0$.
$S \to \infty$, fixed K, τ	$C \sim Se^{-q\tau} - Ke^{-r\tau}$; $P \to 0$; $\Delta_C \to e^{-q\tau}$, $\Delta_P \to 0$.	$\Gamma, \mathscr{V} \to 0$.
$\tau \to \infty$, rates/carry fixed	No universal moneyness-only Greek arrow; exponential discount factors and forward drift matter.	Use full d_1, d_2 instead of naive $\sqrt{\tau}$ power rules.

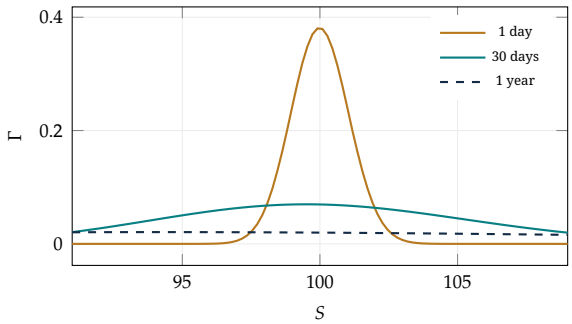
A.9 Where do gamma and vega peak? Conditions matter

Varying strike K with S, τ, σ, r, q fixed: Γ and $\mathscr{V}$ peak at $d_1 = 0$, since their prefactors do not depend on K.

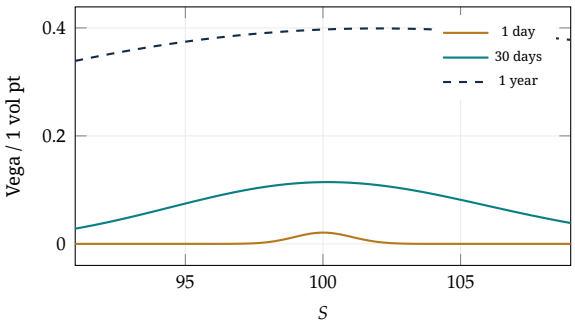
Varying spot S with K, τ, σ, r, q fixed: Γ peaks where $d_1 = -\sigma\sqrt{\tau}$; $\mathscr{V}$ peaks where $d_1 = +\sigma\sqrt{\tau}$.

For example $S^*_\Gamma = K\exp[-(r - q + 3\sigma^2/2)\tau]$ and $S^*_{\mathscr{V}} = K\exp[-(r - q - \sigma^2/2)\tau]$. Thus "both peak *near* ATM" is right, but "both peak exactly at $S = K$" is not always right.

A.10 Quick plots: maturity and moneyness at a glance



Gamma is sharply concentrated near strike at short tenors.



Vega is generally larger at longer maturities near the money.

Parameters: $K = 100$, $r = q = 0$, $\sigma = 20\%$. These are exact BSM gamma and vega curves.

A.11 Futures options: Black–76 in one glance

The ETF-options BSM equations use the ETF's traded share price S. For a *European option on a futures contract*, quote the futures price F and use the discounted Black–76 formulation:

$$\tilde{d}_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad \tilde{d}_2 = \tilde{d}_1 - \sigma\sqrt{\tau},$$

$$C_F = e^{-r\tau}\left[F\Phi(\tilde{d}_1) - K\Phi(\tilde{d}_2)\right], \qquad P_F = e^{-r\tau}\left[K\Phi(-\tilde{d}_2) - F\Phi(-\tilde{d}_1)\right],$$

$$\Delta_F^C = e^{-r\tau}\Phi(\tilde{d}_1), \qquad \Delta_F^P = -e^{-r\tau}\Phi(-\tilde{d}_1),$$

$$\Gamma_F = \frac{e^{-r\tau}\varphi(\tilde{d}_1)}{F\sigma\sqrt{\tau}}, \qquad \mathscr{V}_F = e^{-r\tau}F\varphi(\tilde{d}_1)\sqrt{\tau}.$$

Do not conflate: Black–76 *futures delta* $\partial V/\partial F$ and BSM *spot delta* $\partial V/\partial S$. Futures F itself may depend on the spot via carry; its value is not the futures contract's paid cash premium. Equity/ETF options can have American exercise and discrete distributions, requiring pricing adjustments.

A.12 Cross-check identities and a compact numerical calibration

$$\Delta_C - \Delta_P = e^{-q\tau}, \qquad \Gamma_C - \Gamma_P = 0, \qquad \mathscr{V}_C - \mathscr{V}_P = 0,$$

$$\Theta_C - \Theta_P = qSe^{-q\tau} - rKe^{-r\tau}, \qquad \rho_C - \rho_P = K\tau e^{-r\tau}.$$

For $S = K = 100$, $\sigma = 20\%$, $r = q = 0$, $\tau = D/365$, formulas evaluate to:

Days left	Call Δ	Γ	Vega / +1 pt	Θ/day
1	0.5021	0.3811	\$0.0209	-\$0.2088
30	0.5114	0.0695	\$0.1143	-\$0.0381
365	0.5398	0.0198	\$0.3970	-\$0.0109

A.13 Hedging, actual smile response, and risk P&L

$$\mathrm{d}V \approx \Delta\,\mathrm{d}S + \frac{1}{2}\Gamma(\mathrm{d}S)^2 + \mathscr{V}\,\mathrm{d}\sigma + \Theta\,\mathrm{d}t + \rho\,\mathrm{d}r \quad \text{(finite-move approximation)},$$

$$\Delta_{\text{total}} = \Delta_{\text{BS}} + \mathscr{V}\,\frac{\mathrm{d}\sigma_{\text{imp}}}{\mathrm{d}S}, \qquad \text{(smile/surface convention specified)},$$

$$\mathrm{d}\Delta \approx \Gamma\,\mathrm{d}S + \text{Vanna}\,\mathrm{d}\sigma + \text{Charm}\,\mathrm{d}t,$$

$$\text{PnL}_{\text{hedged}} \approx \frac{1}{2}\Gamma S^2\left(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2\right)\mathrm{d}t \qquad (r = q = 0, \text{ ideal continuous hedge}).$$

Dealer position	Spot moves up	Spot moves down
Long gamma, delta hedged	Sell stock/futures to reduce positive portfolio delta	Buy stock/futures as delta decreases; favorable re-hedge direction
Short gamma, delta hedged	Buy stock/futures to repair negative delta	Sell stock/futures to repair positive delta; adverse re-hedge direction
Long vega	Gains if IV rises, *holding other risk variables fixed*	May lose if IV falls; do not confuse vega with delta
Short vega	Loses if IV rises, *holding other risk variables fixed*	May gain if IV falls; model and smile risks remain

A.14 Fast interview traps and decision checklist

Tempting but wrong shortcut	Precise replacement
"ATM call delta always equals 0.5."	At spot ATM $r = q = 0$, $\Delta_C = \Phi(\sigma\sqrt{\tau}/2) > 0.5$, with limit 0.5 as $\tau \to 0$.
"Delta is exercise probability."	Risk-neutral ITM probability $\Phi(d_2)$, while zero-dividend spot delta is $\Phi(d_1)$.
"IV makes delta approach 0.5."	Vanna sign is $-\operatorname{sign}(d_2)$, and for $\sigma \to \infty$, call delta actually tends $e^{-q\tau}$.
"All Greeks grow if IV grows."	Price rises because vega > 0; Delta, gamma, *and vega itself* may move either way.
"Gamma must grow as maturity shrinks."	True in the *spot-ATM near-expiry limit*; for fixed non-ATM spot, gamma tends to zero.
"Positive theta is guaranteed profit."	Time decay trades against gamma, jumps, IV repricing, hedging costs and adverse selection.
"Call and put have opposite gamma."	Matched European long call and long put have *equal positive* gamma and vega.
"Greek sign is the same for long and short."	Multiply every Greek and P&L by the *signed* contract count and contract multiplier.
"A real price jump changes only delta."	Gamma, vanna and charm drive changing delta; IV surface dynamics can move total delta independently.
"BSM covers every ETF option."	European BSM ignores American exercise, discrete dividends, smiles and underlying-futures versus ETF price distinctions.

Ten-second interview algorithm (1) Call/put and long/short? (2) ITM/ATM/OTM? (3) Spot or forward ATM? (4) Which input changes while what remains fixed? (5) Is it a partial Greek or total surface sensitivity? (6) Which leading formula, sign condition, or expiry limit applies?

A.15 Final mental model

Price is a surface; delta is its spot slope; gamma is curvature; vega measures changes in the surface with implied volatility; theta tracks passage of calendar time; rho is the interest-rate dimension. In a dealer's book these become actionable hedge quantities, not merely mathematical names. Always pair a Greek sign with its position sign, measurement unit, moneyness, and model assumptions.

Model scope. All formulas above concern matched European vanilla options under constant-parameter BSM. American exercise, discrete ETF dividends, dynamic volatility smiles, jumps, stochastic volatility, and hedging costs require model-appropriate adjustments.

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I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

Explore My Worlds

LinkedIn
Original music

GitHub
Open-source projects and research

Spotify
Original music

Instagram
The creative side, on and off stage

Thank you for reading.